

Lec 7-2, Oct 22, Wed .

- Substitution methods (5.5, 5.6, also known as change of variable)
- even and odd functions .
- Volumes using cross sections (6.1)

5.5.6

THEOREM ~~5~~—**The Substitution Rule** If $u = g(x)$ is a differentiable function whose range is an interval I , and f is continuous on I , then

$$\begin{aligned} \int f(g(x)) \underbrace{g'(x) dx}_{d(g(x))} &= \int f(u) du \\ &\stackrel{u=g(x)}{=} \int f(u) du \Big|_{u=g(x)} \end{aligned}$$

Pf of Substitution Rule.

- since f is continuous on I , we have an antiderivative F for f .
- For any x in the domain of g ,

$$f(g(x))g'(x) = F'(g(x))g'(x) = [F(g(x))]'$$

$$\int f(g(x))g'(x) dx = \int [F(g(x))]' dx = F(g(x)) + C.$$

$$\int f(u) du \Big|_{u=g(x)} = F(u) \Big|_{u=g(x)} + C = F(g(x)) + C,$$

which means Left-Hand-Side = Right-Hand-Side. $\square$.

In general, the substitution method seeks to reverse the chain rule. It is usually more convenient to write in the form of indefinite integrals.

$$\begin{aligned} \int f(g(x))g'(x) dx &= \int f(g(x)) d(g(x)) \\ &= \int f(u) du \Big|_{u=g(x)} \\ &= F(u) \Big|_{u=g(x)} + C = F(g(x)) + C. \end{aligned}$$

e.g.2. If f satisfies the condition in the substitution rule, then

$$\begin{aligned}\int f(Ax+B) dx &\stackrel{\substack{u=Ax+B \\ x=\frac{u-B}{A}}}{=} \int f(u) d\left(\frac{u-B}{A}\right) \Big|_{u=Ax+B} \\ &= \int f(u) \frac{1}{A} du \Big|_{u=Ax+B} \\ &= \frac{1}{A} F(u) \Big|_{u=Ax+B} + C \\ &= \frac{1}{A} F(Ax+B) + C.\end{aligned}$$

$\left(\frac{u-B}{A}\right)' = \frac{1}{A}$
 $\Rightarrow d\left(\frac{u-B}{A}\right) = \frac{1}{A} du$

□

e.g.3. $\int \cos(7x+3) dx$

Sol': LHS $\stackrel{\substack{u=7x+3 \\ x=\frac{u-3}{7}}}{=} \int \cos u d\left(\frac{u-3}{7}\right) \Big|_{u=7x+3}$

$$\begin{aligned}&= \int \cos u \frac{1}{7} du \Big|_{u=7x+3} \\ &= \frac{1}{7} \sin u \Big|_{u=7x+3} + C = \frac{1}{7} \sin(7x+3) + C.\end{aligned}$$

□

$\int \sec^2(5x+1) dx$

Sol': LHS $\stackrel{\substack{u=5x+1 \\ x=\frac{u-1}{5}}}{=} \int \sec^2 u d\left(\frac{u-1}{5}\right) \Big|_{u=5x+1}$

$$\begin{aligned}&= \frac{1}{5} \int \sec^2 u du \Big|_{u=5x+1} \\ &= \frac{1}{5} \tan u \Big|_{u=5x+1} + C \\ &= \frac{1}{5} \tan(5x+1) + C.\end{aligned}$$

□

e.g. 4. $\int x \sqrt{2x+1} dx \stackrel{\substack{u=2x+1 \\ x=\frac{u-1}{2}}}{=} \int \frac{u-1}{2} \sqrt{u} d\left(\frac{u-1}{2}\right) \Big|_{u=2x+1}$

$\int x \sqrt{2x+1} dx$
 $= \int \sqrt{2x+1} x dx$
 $= \int \sqrt{2x+1} \frac{1}{2} d(x^2)$
 $\stackrel{\substack{u=x^2 \\ x=\sqrt{u}}}{=} \int \sqrt{2\sqrt{u}+1} \frac{1}{2} du$

$= \frac{1}{4} \int (u-1) \sqrt{u} du \Big|_{u=2x+1}$
 $= \frac{1}{4} \int (u^{\frac{3}{2}} - u^{\frac{1}{2}}) du \Big|_{u=2x+1}$
 $= \frac{1}{4} \left[\frac{2}{5} u^{\frac{5}{2}} - \frac{2}{3} u^{\frac{3}{2}} \right] \Big|_{u=2x+1} + C$
 $= \frac{1}{4} \left(\frac{2}{5} (2x+1)^{\frac{5}{2}} - \frac{2}{3} (2x+1)^{\frac{3}{2}} \right) + C$

□

• Substitution rule in definite integrals.

The rule above can help compute $\int_a^b f(x) dx$.

e.g. 5. $\int_{-1}^2 \underbrace{3x^2}_{u=x^3} \sqrt{\underbrace{x^3+1}_{u+1}} \underbrace{dx}_{du} = \int_{-1}^2 \sqrt{x^3+1} d(x^3)$

$\stackrel{\substack{u=x^3 \\ x=-1}}{=} \int_{-1}^8 \sqrt{u+1} du$

$\stackrel{\substack{y=u+1 \\ y=9}}{=} \int_0^9 \sqrt{y} d(y-1)$

$= \frac{2}{3} y^{\frac{3}{2}} \Big|_0^9 = \frac{2}{3} \cdot 9^{\frac{3}{2}} = 18. \quad \square$

5.6.7

THEOREM 7—Substitution in Definite Integrals If g' is continuous on the interval $[a, b]$ and f is continuous on the range of $g(x) = u$, then

$$\begin{aligned} \int_a^b f(g(x)) \cdot \underline{g'(x)} \, dx &= \int_{g(a)}^{g(b)} f(u) \, du = F(g(x)) \Big|_a^b \\ &= \int_a^b f(g(x)) \, d(\underline{g(x)}) \stackrel{u=g(x)}{=} \int_{g(a)}^{g(b)} f(u) \, du = F \Big|_{g(a)}^{g(b)}. \end{aligned}$$

Pf: Step 1: $\because f$ is continuous $\therefore$ We have an antiderivative F of f .

$$\begin{aligned} \text{RHS} &= \int_{g(a)}^{g(b)} f(u) \, du \stackrel{\text{FTC 2}}{=} F(u) \Big|_{g(a)}^{g(b)} \\ &= F(g(b)) - F(g(a)). \end{aligned}$$

Step 2. Since f is continuous and g' is also continuous, we have $f(g(x)) \cdot g'(x)$ is also continuous and have

$$\begin{aligned} f(g(x))g'(x) &= F'(g(x)) \cdot g'(x) = [F \circ g(x)]' \\ \therefore \int_a^b f(g(x))g'(x) \, dx &= \int_a^b [F \circ g(x)]' \, dx \\ &\stackrel{\text{FTC 2}}{=} F \circ g(x) \Big|_a^b \\ &= F(g(b)) - F(g(a)). \end{aligned}$$

Step 3. LHS = RHS = $F(g(b)) - F(g(a))$. □.

e.g. 6. $\int_0^{\pi/4} (\sin 2x - 2\sin^2 x \sin 2x) dx$

$$= \int_0^{\pi/4} \sin 2x (1 - 2\sin^2 x) dx$$

$$= \int_0^{\pi/4} \sin 2x \cdot \cos 2x dx$$

$$\stackrel{u=2x}{=} \int_0^{\pi/2} \sin u \cdot \cos u d\left(\frac{u}{2}\right)$$

$$= \frac{1}{2} \int_0^{\pi/2} \sin u \cos u du$$

$$= \frac{1}{2} \int_0^{\pi/2} \sin u d(\sin u)$$

$$\stackrel{y=\sin u}{=} \frac{1}{2} \int_0^1 y dy$$

$$= \frac{1}{4} y^2 \Big|_0^1 = \frac{1}{4} \quad \square$$

In-class discussion

e.g. 7. Compute $\int \sin x \cos x dx$ using: $\left(= -\frac{1}{4}(\cos^2 x - \sin^2 x) + C \right)$

1. Double-angle formula: $LHS = \int \frac{1}{2} \sin 2x dx = -\frac{1}{4} \cos 2x + C$

2. Substitution with $u = \sin x$: $LHS = \int \sin x d(\sin x) \stackrel{u=\sin x}{=} \int u du \Big|_{u=\sin x} = \frac{1}{2} u^2 \Big|_{u=\sin x} + C$

3. Substitution with $u = \cos x$: $LHS = \int (-\sin x)(-\cos x) dx = \frac{1}{2} \sin^2 x + C$

$$= \int -\cos x d(\cos x)$$

$$\stackrel{u=\cos x}{=} \int -u du \Big|_{u=\cos x} = -\frac{1}{2} u^2 \Big|_{u=\cos x} + C$$

Rmk. Substitution is not unique.

$$= -\frac{1}{2} \cos^2 x + C$$

$\square$

Method II

$$LHS = \int_0^{\pi/4} \sin 2x dx$$

$$\left(-2 \int_0^{\pi/4} \sin^2 x \sin 2x dx \right)$$

$$= -\frac{1}{2} \cos 2x \Big|_0^{\pi/4}$$

$$- \int_0^{\pi/4} (-\cos 2x + 1) \sin 2x dx$$

$$= \dots$$

$$= \frac{1}{2} \sin^2 x - \frac{1}{4} + C$$

$$= -\frac{1}{4}(1 - \sin^2 x) + C$$

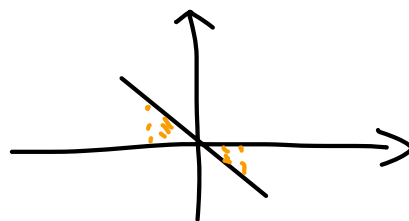
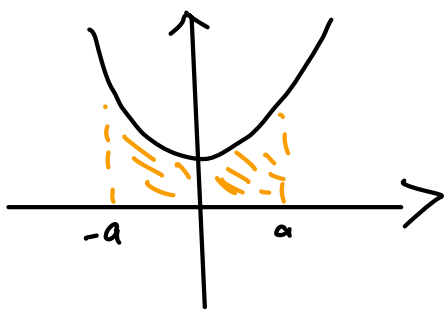
• Even and odd functions

Def: A function $f: D \rightarrow \mathbb{R}$ is called

• an even function, if $f(x) = f(-x)$ for all $x \in D$.

• an odd function, if $f(x) = -f(-x)$ for all $x \in D$.

[This implies that D is symmetric].



Theorem [Integrals of symmetric functions]

Let $f: [-a, a] \rightarrow \mathbb{R}$ be an integrable function.

► If f is an even function, then $\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$.

► If f is an odd function, then $\int_{-a}^a f(x) dx = 0$.

Before proving the theorem, let us see two examples.

eg. 8. $\int_{-\sqrt{2}}^{\sqrt{2}} (15x^4 - 4x^3 + 6x^2 + 7x) dx$

$$= \int_{-\sqrt{2}}^{\sqrt{2}} (15x^4 + 6x^2) dx + \int_{-\sqrt{2}}^{\sqrt{2}} (-4x^3 + 7x) dx$$

$$= 2 \int_0^{\sqrt{2}} (15x^4 + 6x^2) dx + 0$$

$$= 2 \cdot \left(\frac{15}{5} x^5 + \frac{6}{3} x^3 \right) \Big|_0^{\sqrt{2}}$$

$$= 32\sqrt{2} .$$

□.

e.g. 9 $\int_{-1}^3 (x+1)^2 (x-3)^2 dx$

Sol': Observe the symmetry of the integrand about $x=2$!

$$I = \int_{-1}^3 (x-1+2)^2 (x-1-2)^2 dx$$

$$\underline{u=x-1} \quad \int_{-2}^2 (u+2)^2 \cdot (u-2)^2 du$$

$\swarrow \quad \nwarrow$
 $3-1 \quad -1-1$

$$= \int_{-2}^2 (u+2)^2 (u-2)^2 du$$

$$= \int_{-2}^2 (u^2-4)^2 du$$

$$= \int_{-2}^2 [u^4 - 8u^2 + 16] du$$

$$= 2 \int_0^2 [u^4 - 8u^2 + 16] du$$

$$= 2 \left(\frac{1}{5} u^5 \Big|_0^2 - 8 \cdot \frac{1}{3} u^3 \Big|_0^2 + 16 \cdot 2 \right)$$

$$= 2 \left(\frac{1}{5} 2^5 - 8 \cdot \frac{8}{3} + 16 \cdot 2 \right)$$

$$= \frac{512}{15}.$$

□

$$\text{Pt: } \int_{-a}^a f(x) dx = \underbrace{\int_0^a f(x) dx}_{I_1} + \underbrace{\int_{-a}^0 f(x) dx}_{I_2} .$$

(1) If f is even, then

$$I_2 = \int_{-a}^0 f(x) dx$$

$$\underline{\underline{f \text{ even}}} \int_{-a}^0 f(-x) dx$$

$$\underline{\underline{u = -x}} \int_a^0 f(u) d(-u)$$

$$= - \int_a^0 f(u) du$$

$$= \int_0^a f(u) du = I_1$$

$$\Rightarrow \text{LHS} = 2I_1 .$$

$$\therefore \int_{-a}^a f(x) dx = 2I_1 .$$

(2) If f is odd, then

$$I_2 = \int_{-a}^0 f(x) dx$$

$$\underline{\underline{f \text{ odd}}} \int_{-a}^0 -f(-x) dx$$

$$\underline{\underline{u = -x}} \int_a^0 -f(u) d(-u)$$

$$= \int_a^0 f(u) du$$

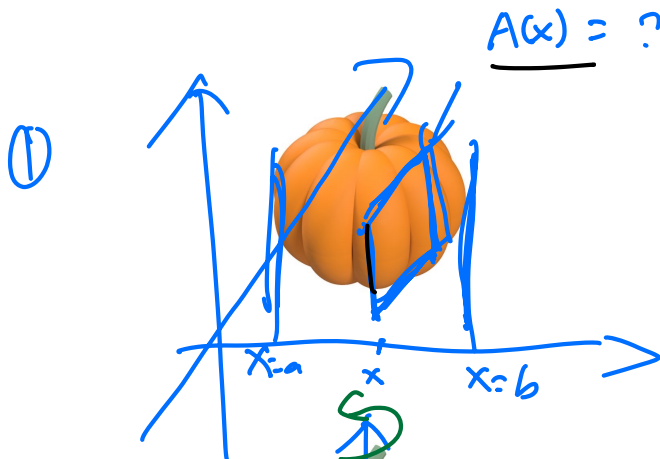
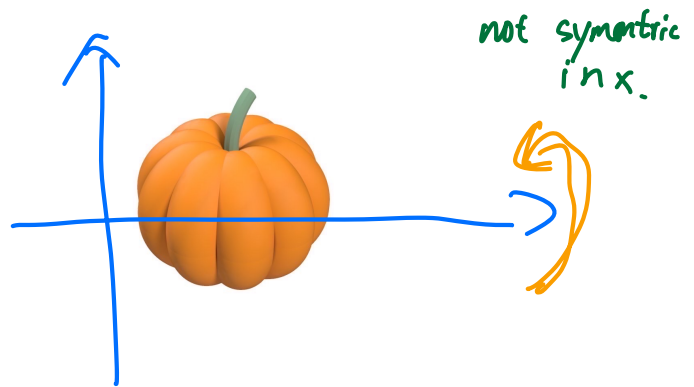
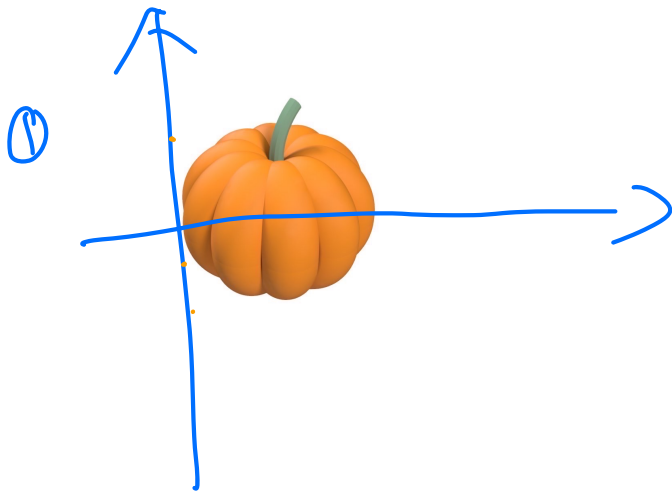
$$= - \int_0^a f(u) du = -I_1 .$$

$$\therefore \int_{-a}^a f(u) du = I_1 + (-I_1) = 0 .$$

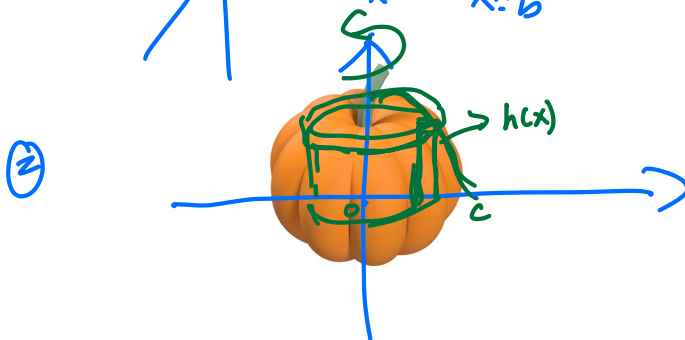
Q.E.D.

Chap 8.

- Volumes using cross-sections (6.1)
- Solid of Revolution (6.1)
- Cylindrical shells (6.2)
- Arc length (6.3)
- Areas of surfaces of revolution (6.4)



← What is the volume?

$$V = \int_a^b A(x) dx$$


$$V = \int_0^c \underline{2\pi x} \cdot \underline{h(x)} dx. \text{ (symmetric)}$$

Volumes Using Cross-Sections

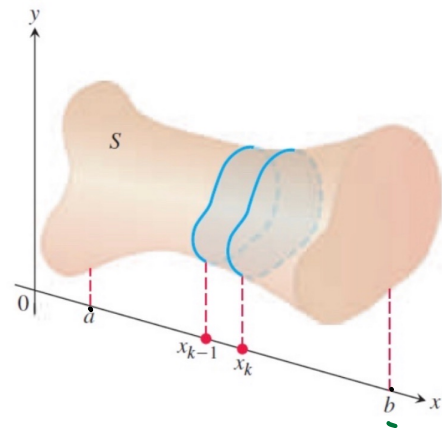
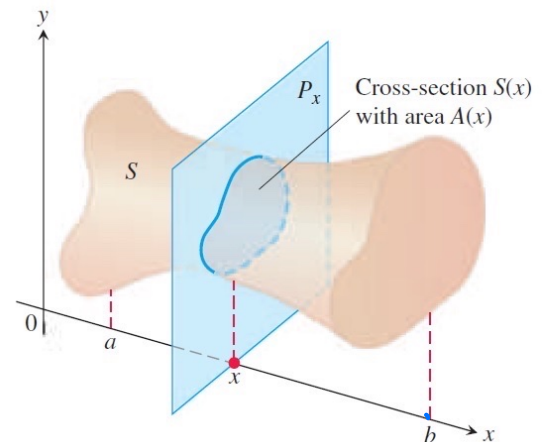
Let S be a solid in the three dimensional Euclidean space (the xyz -space), lying between the planes $x = a$ and $x = b$. How do we compute the volume of the solid?

For $c \in [a, b]$, let $A(c)$ be the area of the cross-section obtained by intersecting S with the plane $x = c$.

- Consider a partition $P := \{x_0, x_1, \dots, x_n\}$ of $[a, b]$.
- When Δx_k is small, the volume of the solid lying between $x = x_{k-1}$ and $x = x_k$ is approximately $A(x_k) \Delta x_k$.
- When $\|P\|$ is small, the volume of S is approximately

$$\sum_{k=1}^n A(x_k) \Delta x_k.$$

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n A(x_k) \cdot \Delta x_k = \int_a^b A(x) dx.$$



Definition

Let S be a solid that lies between the planes $x = a$ and $x = b$. The **volume** V of S is defined by

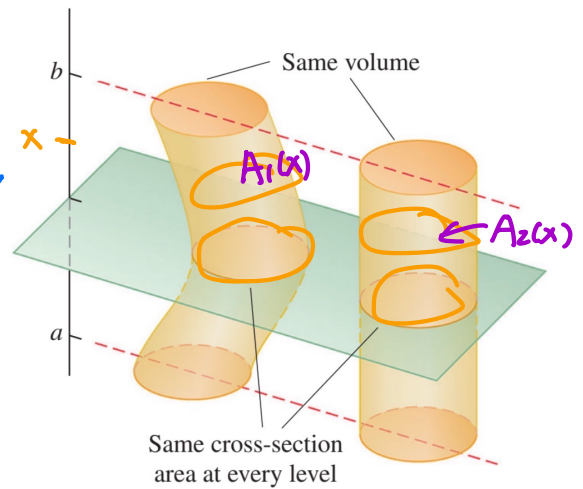
$$V := \int_a^b A(x) dx,$$

provided that the cross-section area function $A(x)$ is integrable.

• To find V , we need to find a formula for the cross section Area $A(x)$ first.

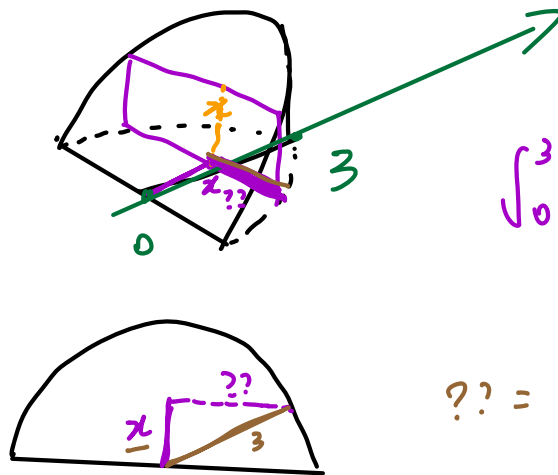
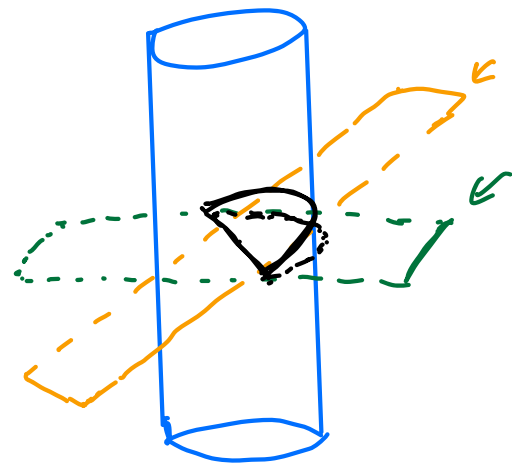
If $A_1(x) = A_2(x)$, $\forall x \in [a, b]$,

then $V_1 = V_2$!



EXAMPLE 6.1.2

A curved wedge is cut from a circular cylinder of radius 3 by two planes. One plane is perpendicular to the axis of the cylinder. The second plane crosses the first plane at a 45° angle at the center of the cylinder. Find the volume of the wedge.



$$\int_0^3 A(x) dx$$

$$?? = \sqrt{9-x^2}$$

$$A(x) = 2 \cdot x \cdot ??$$

$$= 2 \cdot x \cdot \sqrt{9-x^2},$$

$$x \in [0, 3].$$

Sol': $A(x) = 2 \sqrt{9-x^2} x$, $x \in [0, 3]$.

(check: $A(0) = 0$, $A(3) = 0$)

$$\therefore V = \int_0^3 A(x) dx = \int_0^3 2 \sqrt{9-x^2} x dx = \int_0^3 \sqrt{9-x^2} d(x^2)$$

$$\stackrel{u=x^2}{=} \int_0^9 \sqrt{9-u} du$$

$$= -\frac{2}{3} (9-u)^{\frac{3}{2}} \Big|_0^9 = 18 \quad \text{Q.E.D.}$$

Solid of Revolution

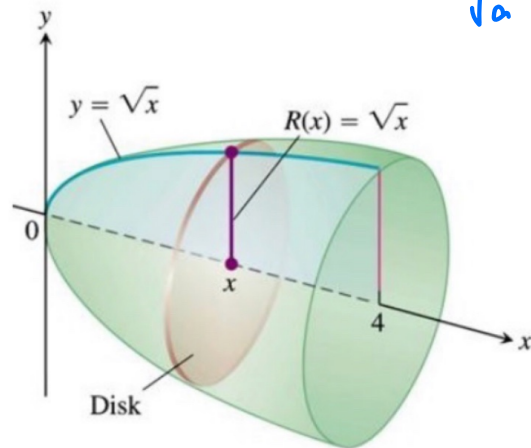
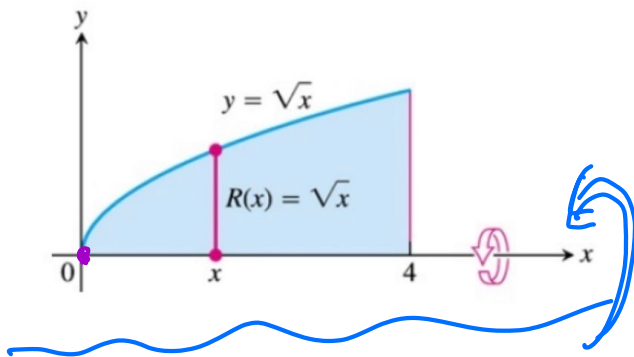
If the solid S is generated by rotating the region

$$\{(x, y) : 0 \leq y \leq R(x), \ a \leq x \leq b\}$$

around the x -axis, then the cross-sections of S are discs with radii $R(x)$. Consequently, $A(x) = \pi R(x)^2$, and

$$V = \int_a^b \pi R(x)^2 dx.$$

$$\begin{aligned} V &= \int_a^b A(x) dx \\ &= \int_a^b \pi R(x)^2 dx \end{aligned}$$



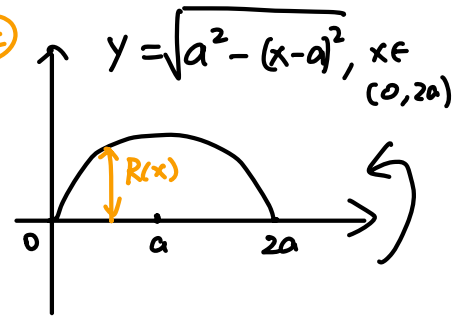
e.g. Derive the volume formula of a sphere with radius a .

Sol: $\dots$

$$\begin{cases} (x-a)^2 + y^2 = a^2 \\ y \geq 0 \end{cases} \Rightarrow y = \sqrt{a^2 - (x-a)^2}, \quad x \in (0, 2a)$$

Area is given by

$$\begin{aligned} \{(x, y) : 0 \leq x \leq 2a, \ 0 \leq y \leq \sqrt{a^2 - (x-a)^2}\} \\ = \sqrt{-x^2 + 2ax} \end{aligned}$$



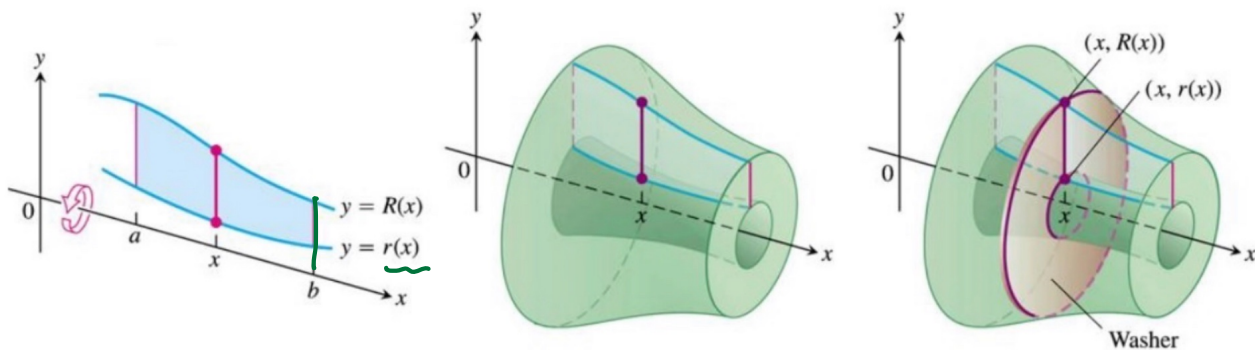
$$\begin{aligned}
 \therefore V &= \int_0^{2a} \pi \cdot \left(\sqrt{-x^2 + 2ax} \right)^2 dx \\
 &= \int_0^{2a} \pi (-x^2 + 2ax) dx \\
 &= \pi \left(-\frac{1}{3}x^3 \right) \Big|_0^{2a} + 2a\pi \frac{1}{2}x^2 \Big|_0^{2a} \\
 &= \pi \left(-\frac{1}{3}8a^3 \right) + a\pi \cdot 4a \\
 &= \frac{4}{3}\pi a^3. \quad \square
 \end{aligned}$$

If the solid S is generated by rotating the region

$$\{(x, y) : 0 \leq r(x) \leq y \leq R(x), a \leq x \leq b\}$$

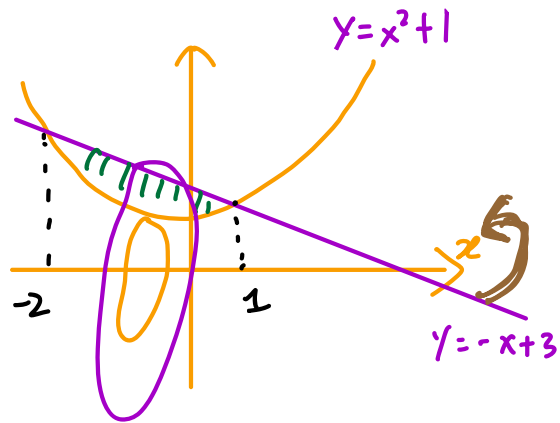
around the x -axis, then similarly,

$$V = \int_a^b \pi \left(R(x)^2 - r(x)^2 \right) dx. \quad A(x) = \pi R(x)^2 - \pi r(x)^2$$



6.1.9

EXAMPLE The region bounded by the curve $y = x^2 + 1$ and the line $y = -x + 3$ is revolved about the x -axis to generate a solid. Find the volume of the solid.



Sol' : Let $x^2 + 1 = -x + 3$

$\Rightarrow x_1 = -2$ and $x_2 = 1$.

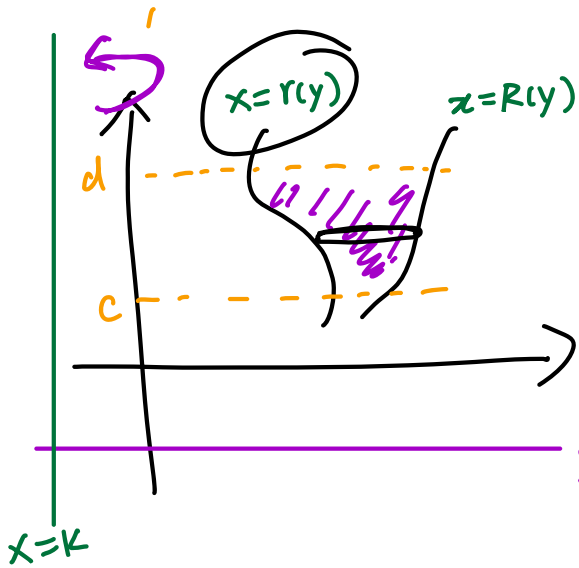
$\therefore$ Area is given by

$$\{(x, y) : -2 \leq x \leq 1, \underbrace{x^2 + 1}_{r(x)} \leq y \leq \underbrace{-x + 3}_{R(x)}\}$$

$$\begin{aligned} \therefore V &= \int_{-2}^1 \pi (R(x)^2 - r(x)^2) dx \\ &= \int_{-2}^1 \pi ((-x + 3)^2 - (x^2 + 1)^2) dx \\ &= \frac{117}{5} \pi. \end{aligned}$$

□

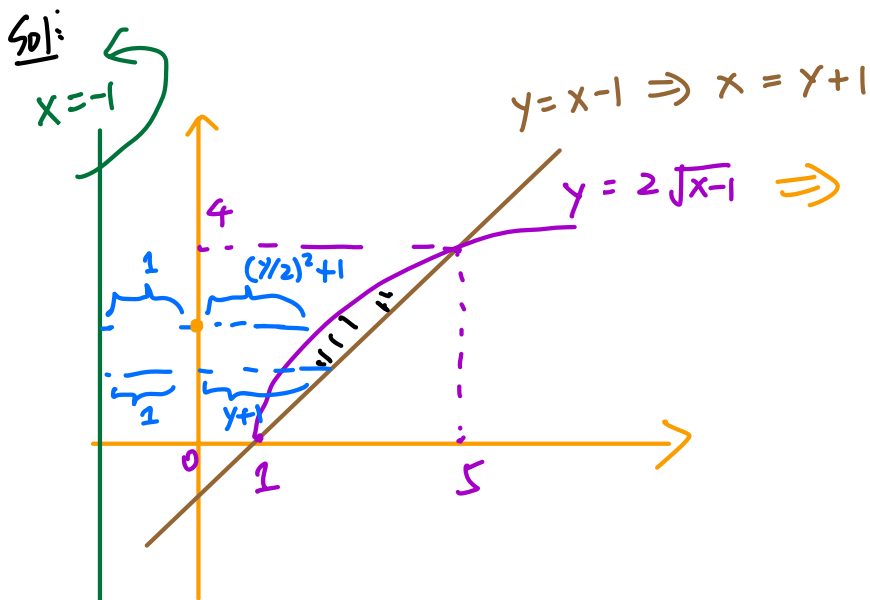
The ideas above can be applied similarly to a solid obtained by revolving a region around the y -axis, or around lines of the form $y=k$ or $x=k$ in general.



$$\textcircled{1} \quad \text{Region} = \left\{ (x, y) : c \leq y \leq d, \right. \\ \left. r(y) \leq x \leq R(y) \right\}$$

$$\Rightarrow V = \int_c^d \pi (R(y)^2 - r(y)^2) dy$$

e.g. Find the volume of the solid obtained by rotating the region bounded by $y = \sqrt{x-1} \cdot 2$ and $y = x-1$ about $x = -1$.



$$y = x-1 \Rightarrow x = y+1$$

$$y = 2\sqrt{x-1} \Rightarrow x-1 = \left(\frac{y}{2}\right)^2 \Rightarrow x = 1 + \left(\frac{y}{2}\right)^2$$

$$x-1 = 2\sqrt{x-1}$$

$$(x-1)^2 = 4(x-1)$$

$$(x-1)(x-1-4) = 0$$

$$\Rightarrow x_1 = 1 \text{ and } x_2 = 5$$

$$\cdot \text{Region} = \left\{ (x, y) : 0 \leq y \leq 4, \right. \\ \left. 0 \leq \left(\frac{y}{2}\right)^2 + 1 \leq x \leq y+1 \right\}$$

$$\cdot V = \int_0^4 \pi \left[(y+1) - (-1) \right]^2 - \left[\left(\frac{y}{2}\right)^2 + 1 - (-1) \right]^2 dy$$

$$=$$

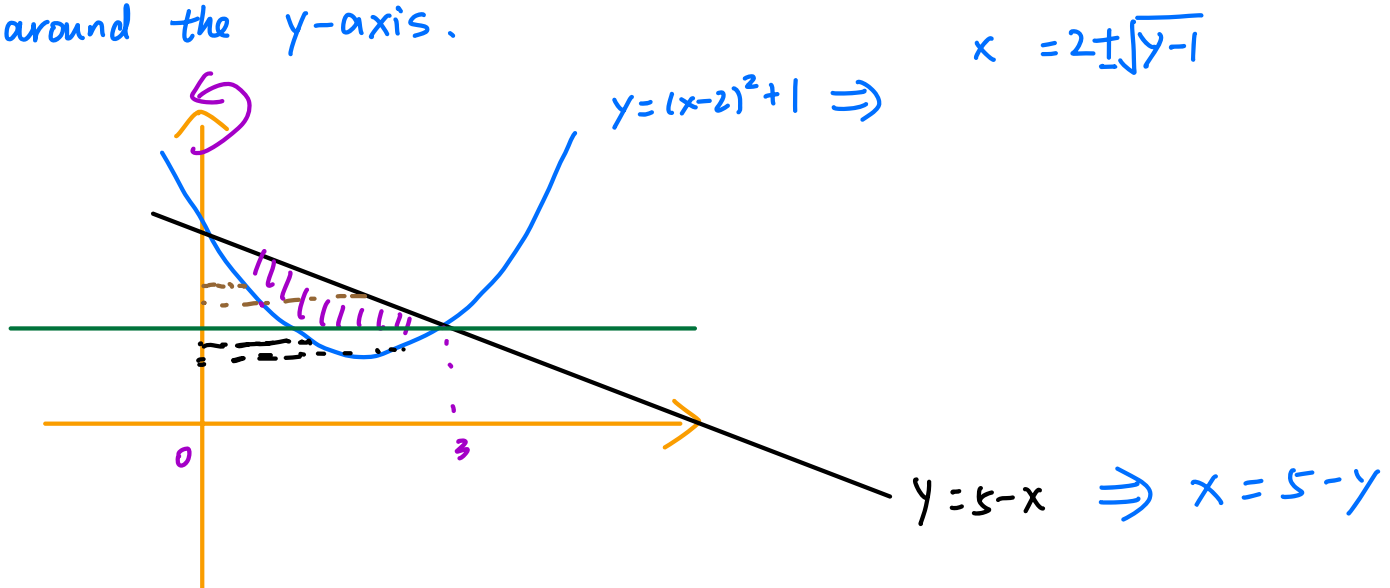
$$= \int_0^4 \pi \left(4y - \frac{1}{16} y^4 \right) dy$$

$$= \frac{96}{5} \pi.$$

图.

e.g. A region is bounded by $y = (x-2)^2 + 1$ and $y = 5-x$.

Find the volume of the solid generated by revolving the region around the y -axis.



Sol': Step 1. intersection!

Let $(x-2)^2 + 1 = 5-x \Rightarrow x_1 = 0$ and $x_2 = 3$

Step 2: write down the region.

$$\text{Region} = \{ (x, y) : 0 \leq x \leq 3, (x-2)^2 + 1 \leq y \leq 5-x \}$$

$$= \{ (x, y) : \underbrace{2-\sqrt{y-1}}_{r_2(y)} \leq x \leq \underbrace{2+\sqrt{y-1}}_{R_2(y)}, 1 \leq y \leq 2$$

$$\text{and } \underbrace{2-\sqrt{y-1}}_{r_1(y)} \leq x \leq \underbrace{5-y}_{R_1(y)}, 2 \leq y \leq 5 \}$$

Step 3: cross sections and volume.

$$\begin{aligned} \therefore V &= \int_1^2 \pi (R_2(y)^2 - r_2(y)^2) dy + \int_2^5 \pi (R_1(y)^2 - r_1(y)^2) dy \\ &= \int_1^2 \pi ((2+\sqrt{y-1})^2 - (2-\sqrt{y-1})^2) dy + \int_2^5 \pi ((5-y)^2 - (2-\sqrt{y-1})^2) dy \\ &= \pi \left[\int_1^2 (4 + y - 1 + 4\sqrt{y-1}) - (4 - 4\sqrt{y-1} + y - 1) dy \right. \\ &\quad \left. + \int_2^5 (25 + y^2 - 10y - (4 - 4\sqrt{y-1} + y - 1)) dy \right] \end{aligned}$$

= ugly computation...

$$= \frac{27}{2} \pi.$$

□.